ES 1C03 · ENVIRONMENTAL SCIENCE

Fast Fashion: *Environmental Impact* in Canada

THE HIDDEN COST OF CHEAP CLOTHING

Fast fashion — the rapid production of inexpensive, trend-driven clothing — drives a linear "take-make-dispose" model with severe environmental consequences.[1] Canadians purchase an estimated **70 garments per person annually**, yet the average item is worn only **7–10 times**.[2,3] The fashion industry accounts for **8–10% of global carbon emissions** — more than international flights and shipping combined.[5]

01 The Fast Fashion Lifecycle

RAW MATERIALS ▸ MANUFACTURING ▸ TRANSPORT ▸ CONSUMER USE ▸ DISPOSAL

02 Key Statistics

12 kg
Textile waste per Canadian per year
(ECCC, 2023)[1]

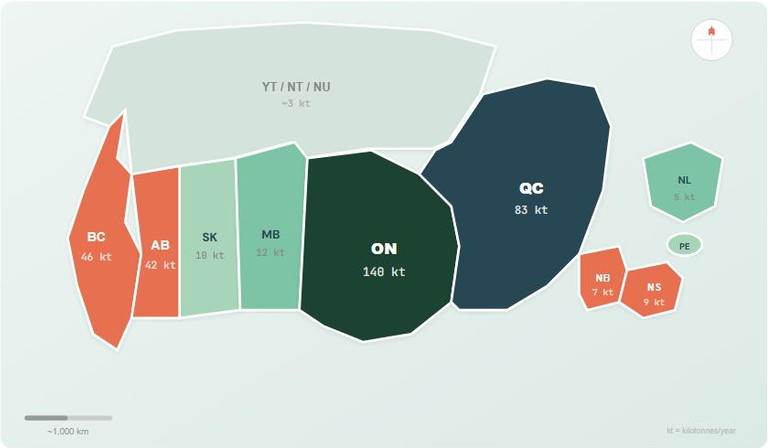
2,700 L
Water to produce one cotton T-shirt
(EMF, 2017)[3]

1.2 Gt
CO_2 from global fashion industry/yr
(UNEP, 2023)[4]

<1%
Clothing recycled into new garments
(EMF, 2017)[3]

03 Data Visualizations

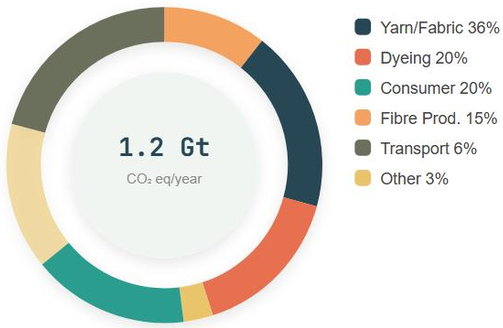
FIGURE 1 · TEXTILE WASTE BY PROVINCE (KILOTONNES/YR)



>100 kt · 30–100 kt · 10–30 kt · <10 kt

Figure 1. Estimated annual textile waste by province, population-weighted. Ontario and Quebec generate the most due to higher populations.[1,2]

FIGURE 2 · FASHION GHG EMISSIONS BY STAGE



One pair of jeans uses ~7,500 L of water to produce[3]

Figure 2. Data: IPCC (2022); Ellen MacArthur Foundation (2017)[3,5]

04 Environmental Impacts in Canada

Carbon Emissions

Fashion produces ~1.2 Gt CO_2/yr globally. Synthetic fibres from fossil fuels generate emissions during production and incineration.[4,5]

Water Pollution

Fashion consumes ~93 billion m³ of water/yr. Textile dyeing is the 2nd-largest water polluter, introducing heavy metals into freshwater.[3,4]

Landfill & Waste

Canadians send ~12 kg of textile waste per person to landfills annually. Synthetics take 20–200 years to decompose.[1,2]

Biodiversity Loss

Cotton uses 6% of global pesticides and 16% of insecticides. Fibre crop expansion drives deforestation globally.[4,5]

ACTION Solutions: How You Can Help

BUY LESS, CHOOSE WELL
Invest in durable garments — reducing purchases by 30% could cut emissions equal to removing 1M cars.[3]

THRIFT & SWAP
Shop second-hand or organize clothing swaps to extend garment lifespans.[2]

RECYCLE & DONATE
Use textile recycling programs — many Canadian cities offer curbside collection.[1]

WASH MINDFULLY
Cold water and microfibre bags reduce microplastic release by up to 80%.[3]

ADVOCATE FOR POLICY
Support extended producer responsibility (EPR) under Canada's Zero Plastic Waste strategy.[1]

SUSTAINABLE FABRICS
Choose organic cotton, TENCEL™, or recycled fibres with GOTS/OEKO-TEX certification.[4]

REFERENCES

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[2] Statistics Canada (2023) *Household spending on clothing and textiles.* Statistics Canada. Available at: https://tinyurl.com/statcan-clothing (Accessed: 20 February 2026).
[3] Ellen MacArthur Foundation (2017) *A new textiles economy: Redesigning fashion's future.* Ellen MacArthur Foundation. Available at: https://tinyurl.com/emf-textiles (Accessed: 20 February 2026).
[4] United Nations Environment Programme (2023) *Sustainability and circularity in the textile value chain.* UNEP. Available at: https://tinyurl.com/unep-textiles (Accessed: 20 February 2026).
[5] Intergovernmental Panel on Climate Change (2022) *Climate Change 2022: Mitigation of Climate Change.* Cambridge University Press. Available at: https://tinyurl.com/ipcc-ar6-wg3 (Accessed: 20 February 2026).
[6] Plastic Collective (2024) *How fast fashion is bad for the environment.* Plastic Collective. Available at: https://www.plasticcollective.co/how-fast-fashion-is-bad-for-the-environment/ (Accessed: 25 February 2026).